

Temporal Associative Memory

Online sequence learning

The “temporal” associative memory component learns higher-order sequences and next-token prediction on the fly.

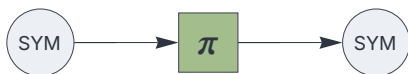
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",
  "hyperparameters": {"default": [2000, 10]},
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},
    {"component": "temporal", "send": [21], "receive": [11], "capacity": 3, "decimate": 0.8},
    {"component": "output", "plugin": "codec", "receive": [21]}
  ]
}
```

```
circ = Circuit[json];
c = circ["function"];
circ["schematics"][ImageSize→220]
```



Dataset

Generate a 2000 character random string.

```
chars = CharacterRange["A", "Z"]
```

```
{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z}
```

```
dataset = RandomChoice[ chars, 2000];
```

```
StringJoin[dataset]
```

```
MFMINYVDTWXYICTQIBKAPILAKLQSBUPQXRSLTMNUNOXZLIJHHDSPPRPAZKYAVBEBFWRXRDZO\
SGOBXOREMNFLHMCZJCDCHBXOVSIHTFKHPKTQCNIVAHFLNCHVQNFXMIVOPHJWOTDLWDCW\
ZREJWXFRPTZBVLVHMMDGYLHIVPVWTTFFVMCQQJJZXJAVEMYFTZSZYHSVGBLUDJECQKUNXQ\
WBGDLBMQSICTCBOQYYFGTWKDDPSHDMKYVAVCHOMCRSFPSCCKFENSGNIBVZAFDOWYGTKHF\
HWPBHQXSDKFVOVSXTGYFKLAPBTNNKOIEVLCSNDBQBTHBUNNUGUJIJFLWPZUFPFPFMKHM\
CHAESBJACUQLNYKJJEOUMWUQKKCTWVOLDUCHDJVNKRWPQKXBNXDBTVWTTYASOYXSWMLEJ\
WUHBXCRHRZESDEFBBAEWFKBIXLELMPPANTJMFNSMMCAYGYUENIBZAQIYCBWKFDEUUVUO\
RNQNBZIIDXUKGUZQFBRUPUEKYCZEZZNOYFVAJKTFCIKWRTFROI OONFOVTKYNTRHQSEPW\
JUDHQXKICOEKLVGWJXPQGQREUDZGXHPYSTHKUSLXTKBGTBAEIMUFYAKNMNHVYEUITCJA\
UDGLDFHLIIIXDDAQIDIDLLJOOWOEASUSZGDLVWDTXEHZETUJITYSMIWMXMNSYIXKRCRNX\
SRSNALUGGKXVSLJV FHCJFWNJLESJCILZCMXMCVOTRENATYXGWRWFYOFGKITKMLEPOAMRCV\
PTESYWYDPZMMCTMBANIYAWUKHVLQIYPRPRIURIDVXJRLCEJMLHXDFFZQKKNRHFUSAPYDE\
NNMAIBGRAIXOOXBXKGPIZKYWCDWVLLFPFRBYKZCOKWOFRRERPMFLQXUXEJORCZMTFNPCSX\
YAHETDBHUZKNNZTUXHYXBNYPBTYPVRWMNPZDVLLSEUQWDDJEQEWITPUFEOPHFUHD BHMYS\
CEZNWRSYFBSQNEAQQRGOOHODDSIKPOVXCYOYTHZVVDVLGZESSOSNHYRJVVKVASILYLFSO\
DBGQLACQUBYYHPXCIDRLBCRESKXFHRCJQNDWOOPMKLNSSUXNOSBFDGILFZWHFUXDAWEVF\
QUOANQGYDSPJNFLTQBPZPNHZRQSBVYXGYRUODINTWANA JFZUDOLNBZHWNTWYEVXGRNKDDA\
NZHPLTDAJUAYETVVVCMGZCZIDVEFEMHAVIYWEKRQBFLSAQNFEFXDTCMMKYM XRXTMTFUIR\
SFTKJTEPBGXFPGVQUQPFWYTCEEXNWQBCRSLAAGLHGTQIGXQRUKSSXHKIDHCFNZIAIQGEX\
KSYXMRCBPCLOUKQBHIBOFDBROPWPFCQSCSKVYZWEHAAULVNCRZOKQUCSJFAVTFRUCCVA\
ZPHMDNLVUBTOMVNNGUYVVRKYGGNEJZZWOFPHMCUVJIZIQXZRAVRZXJZJJJXOYQKZJDLGG\
BILHEKQSBVHEYFQDZH ZEBQXGOKQQXNFUGPSOAIKUVB AEUQGZIVIPETFNPLYTONHHMWB BT\
CQMEGXWACLEDNRVWIYXSSBDKQVRBFZZHDMEL SJMJPYVDWOLPQVTPVBXWFWTCYARRHFXMW\
OMREOWEWGZMKHRLXWOGGPUPWYZTMATSSLYJEFQPLORHQASQSAJXZBDQUURRHWOQHGGBI\
RMADNHXQGGROHBDPEWLUHJPCCJLVQJSFOFRFOCTRKIMJAMITFHJJBHKEVANEIHYWWZSPF\
ZMMWJTPLJCBDDAQOECCBBZGBFMPNDEAOZLXLFWRBFXMGLSVJDLWNRYTVOZY ZANDVISZXD\
UXJFVSPBHAUMGAMYKVHDGUJXRNVSDTVSSWNTENMAOOMGVSWADIXYGKTYVMSMBLKL SXHZO\
JFMFHBHXLHYNXVDRBGWJNHRFCDHTTWBJDVHICIVTMUNZVVQQQEMIVUEBFYCZOXDDVXUGF\
TVSNMWOHPGOBVXVYUDNYYHPKDCHLQQYXWFJMGKSKTVBKAMUYEQCGTSVYAZXBDEBKO
```

Prediction score

```
Do[
  predictions = c /@ dataset;
  Print["  iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
    {i, 1, 10}];
```

iteration = 1, prediction score = 0.0305

iteration = 2, prediction score = 0.87

iteration = 3, prediction score = 0.9805

iteration = 4, prediction score = 0.994

iteration = 5, prediction score = 0.9955

iteration = 6, prediction score = 0.9985

iteration = 7, prediction score = 0.9995

iteration = 8, prediction score = 0.9995

iteration = 9, prediction score = 1.

iteration = 10, prediction score = 1.